

FTN-002 - Foundations of FT, FTI, FTN, and FTC

Four Core Concepts

The Function Tunnel framework is built upon four progressively larger concepts:

FT
↓
FTI
↓
FTN
↓
FTC

FT: Function Tunnel

A Function Tunnel is a stable functional pathway that supports a particular category of perception, reasoning, decision making, or action.

A Function Tunnel is not a piece of knowledge.

Instead, it is a reusable operational structure.

Examples include:

- mathematical reasoning pathways,
- musical perception pathways,
- software debugging pathways,
- scientific hypothesis generation pathways.

FTI: Function Tunnel Intelligence

Function Tunnel Intelligence describes the ability to:

- construct tunnels,
- maintain tunnels,
- repair tunnels,
- connect tunnels,
- and evolve tunnels.

From this perspective, intelligence is not unrestricted optimization.

It is the ability to operate effectively within and across functional pathways.

FTN: Function Tunnel Networks

Individual tunnels rarely exist in isolation.

As tunnels accumulate, they form interconnected networks.

These networks support:

- transfer learning,
- cross-domain reasoning,
- creativity,
- specialization,
- collective coordination.

FTN represents the structural ecosystem of intelligence.

FTC: Function Tunnel Capital

Function Tunnel Capital represents the accumulated value of Function Tunnel Networks.

FTC determines:

- future learning efficiency,
- problem-solving capacity,
- adaptability,
- and long-term comparative advantage.

FTC functions similarly to economic capital, but operates within the domain of intelligence and capability formation.

Knowledge Feeds FT but Is Not FT

One of the central hypotheses of this work is:

Knowledge feeds Function Tunnels, but knowledge is not Function Tunnels.

Knowledge provides material.

Function Tunnels provide structure.

Without tunnels, knowledge often remains isolated information.

With tunnels, knowledge becomes operational capability.

A Structural View of Human Development

Under the FTN framework:

- learning becomes tunnel formation,
- expertise becomes tunnel stabilization,
- creativity becomes tunnel recombination,
- wisdom becomes tunnel network optimization.